

Fall 2025.

**Physics 509. Quantum Theory I.
Yurii Shylnov.**

Homework # 4

Due to October 29th, 2025.

You can use any source to solve the problems. However, your solutions should be presented in a very detailed and logical way. You need to understand every step of your solution and be able to explain it.

1. Use the second quantization formalism to calculate the first order non-linear correction to one-dimensional harmonic oscillator energy levels if the non-linear term is $\hat{\mathcal{H}}_I = \lambda \hat{p}^4/4$.
2. Now use the second quantization formalism to calculate the SECOND order non-linear corrections to one-dimensional harmonic oscillator energy levels if the non-linear term is $\hat{\mathcal{H}}_I = \lambda \hat{x}^3/3$. The first order correction is equal to zero (prove it).
3. Calculate energy levels of a two-dimensional isotropic oscillator with the Hamiltonian of

$$\hat{H}(x, y) = \frac{1}{2m} (\hat{p}_x^2 + \hat{p}_y^2) + \frac{1}{2}k (\hat{x}^2 + y^2) + \alpha \hat{x} \hat{y}.$$

Consider the case of $|\alpha| < k$. What happens if $|\alpha| > k$?

Hints: Diagonalize the potential energy quadratic form. Double check that the Laplace operator from the kinetic energy term is still diagonal in new coordinates. Now, you have a sum of two independent one-dimensional harmonic oscillators.

4. For the oscillator from the previous problem find the first order correction of the ground energy level treating the term $V = \alpha \hat{x} \hat{y}$ as a perturbation.
5. Now, for the same model calculate the first order corrections of a) the first b) the second excited energy levels. Find correct zero order eigenfunctions. Hint: you will need to figure out the degeneracy, matrix elements of the perturbation, and finally solve a secular equation.

6. An external homogeneous time-dependent electric field $\mathcal{E}(t) = \mathcal{E}_0 \exp(-t^2/\tau^2) \cos \omega_0 t$ is applied on a one-dimensional charged harmonic oscillator. If for $t \rightarrow -\infty$ the oscillator is at the n^{th} energy level, what is the probability of its transition to the other energy levels (which ones?) when $t \rightarrow +\infty$?
7. A time-dependent periodic perturbation of $\hat{V}(x, t) = -\hat{x}F_0 \cos \omega_0 t$ is applied on a particle in a δ -well. In the first perturbation order, calculate a probability per time unit of the particle transition to the continuous spectrum (sort of ionization). Hint: The Golden Fermi Rule is applicable here.